

8. (g) Implementation Plan: B.Tech. in Biotechnology

Semester – I							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21LEH101T	Communicative English	2	1	0	3		
21MAB101T	Calculus and Linear Algebra	3	1	0	4		
21PYB101J	Physics: Electromagnetic Theory, Quantum Mechanics, Waves and Optics	3	1	2	5		
1MES102L ¹	Engineering Graphics and Design	0	0	4	2		
21EES101T	Electrical and Electronics Engineering	3	1	0	4		
21CYM101T ¹	Environmental Science*	1	0	0	0		
21PDM101L ¹	Professional Skills and Practices	0	0	2	0		
21LEM101T ¹	Constitution of India	1	0	0	0		
Total Credits					18		
Semester – II							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21LEH102T	Chinese						
21LEH103T	French						
21LEH104T	German						
21LEH105T	Japanese	2	1	0	3		
21LEH106T	Korean						
21LEH107T	Spanish						
21LEH108T	Russian						
21GNH101J	Philosophy of Engineering	1	0	2	2		
21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4		
21CYB101J	Chemistry	3	1	2	5		
21BTB105T	Cell Biology	2	0	0	2		
21CSS101J	Programming for Problem Solving	3	0	2	4		
21BTC101T	Biochemistry	3	0	0	3		
21MES101L ¹	Basic Civil and Mechanical Workshop	0	0	4	2		
21PDM102L ¹	General Aptitude*	0	0	2	0		
21GNM101L ¹	Physical and Mental Health using Yoga						
21GNM102L ¹	National Service Scheme	0	0	2	0		
21GNM103L ¹	National Cadet Corps						
21GNM104L ¹	National Sports Organization						
Total Credits					25		
Semester – III							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21CHS251T	Basic Chemical Engineering	3	0	0	3		
21PDH209T ¹	Social Engineering	2	0	0	2		
21BTC201L ¹	Biochemistry Laboratory	0	0	4	2		
21BTC202T	Microbiology	3	0	0	3		
21BTC203L ¹	Cell and Microbiology Laboratory	0	0	4	2		
21BTC204T	Bioprocess Principles	3	0	0	3		
21BTC205L ¹	Bioprocess Principles Laboratory	0	0	4	2		
21BTC206T	Genetics and Cytogenetics	3	0	0	3		
21LEM201T ¹	Professional Ethics	1	0	0	0		
21PDM201L ¹	Verbal Reasoning	0	0	2	0		
21LEM202T ¹	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	2	1	0	3		
Total Credits					23		
Semester – IV							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21CHS252J	Chemical Engineering Principles	3	0	2	4		
21CSC206T	Artificial Intelligence	3	0	0	3		
21BTC207T	Molecular Biology	3	0	0	3		
21BTC208L ¹	Molecular Biology Laboratory	0	0	4	2		
21BTC209T ²	Bioprocess Engineering	3	0	0	3		
21BTC210L ¹	Bioprocess Engineering Laboratory	0	0	4	2		
E	Professional Elective – I				3		
21DCS201P ¹	Design Thinking and Methodology	1	2	0	3		
21PDM202L ¹	Critical and Creative Thinking Skills	0	0	2	0		
Total Credits					23		
Semester – V							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21MAB303T	Biostatistics for Biotechnologists	3	1	0	4		
21BTC301J	Gene Manipulation and Genomics	3	0	2	4		
21BTC302J	Immunology	3	0	2	4		
21BTC303T	Protein Engineering	3	0	0	3		
E	Professional Elective – II				3		
O	Open Elective – I				3		
21GNP301L ¹	Community Connect	0	0	2	1		
21PDM301L ¹	Analytical and Logical Thinking Skills	0	0	2	0		
21LEM301T ¹	Indian Art Form	1	0	0	0		
Total Credits					22		
Semester – VI							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21CSS303T	Data Science	1	1	0	2		
21BTC304T	Animal Biotechnology	3	0	0	3		
21BTC305L ¹	Animal Biotechnology Laboratory	0	0	4	2		
21BTC306T	Plant Biotechnology	3	0	0	3		
E	Professional Elective – III				3		
O	Open Elective – II				3		
21BTP302L ¹	Project	0	0	6	3		
21BTP303T ¹	MOOC	3	0	0	3		
21LEM302T ¹	Indian Traditional Knowledge	1	0	0	0		
21PDM302L ¹	Employability Skills and Practices	0	0	2	0		
Total Credits					19		
Semester – VII							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21BTC401L ¹	Plant Biotechnology Laboratory	0	0	4	2		
21BTC402J	Bio Separation Technology	3	0	2	4		
E	Professional Elective – IV				3		
E	Professional Elective – V				3		
O	Open Elective – III				3		
21GNH401T	Behavioral Psychology	2	1	0	3		
21BTM191T ¹	Bioethics and IPR	1	0	0	0		
Total Credits					18		
Semester – VIII							
Course Code	Course Title	Hours / Week					
		L	T	P	C		
21BTP401L	Major Project	0	0	30	15		
21BTP402L	Major Project	0	0	20	10		
21BTP403L	Internship#	0	0	10	5		
Total Credits					15		

#Students have to register either 21BTP401L or 21BTP402L and 21BTP403L both in eighth semester